A Hybrid Analytical-Computational Matrix Framework for Decoupling Alkyne Poisoning Channels in Heterogeneous Catalysis

Aniema Ekemini Johnson[1]

[1]Southern British High School, Cross River State, Calabar, Nigeria

Project Descriptometrix Manuscript — June 18, 2026

Abstract

The design of transition-metal catalysts resistant to deactivation by trace contaminants remains a foundational bottleneck in industrial chemical processes. Traditional materials screening methods rely heavily on intensive first-principles density functional theory calculations, which are too computationally demanding for high-throughput discovery workflows. In this work, we present Project Descriptometrix, a multi-layered, hybrid analytical-computational framework designed to isolate and evaluate catalyst poisoning mechanisms in fractions of a millisecond. By projecting the complex quantum mechanical interaction space onto a localized, tridiagonal three-center Hückel-Coulson Hamiltonian matrix, we derive two explicit, non-iterative parameters: backbonding susceptibility (ΔE_π) and Coulson terminal carbon charge redistribution (Δq_{C1}). We validate this engine against an elite material dataset, showing that it cleanly decouples harmless, reversible π-coordination from permanent, destructive σ-acetylide catalyst deactivation. Integrating these physics-informed metrics into a localized machine learning and Kinetic Monte Carlo framework reveals unique scaling-relation outliers—such as spatial confinement cavities (Fe_1–N_4) and dynamic charge-inversion single-atom alloys ($Pd_1/Ag(111)$)—that bypass conventional catalyst performance limits. The complete, dockerized API pipeline is made open-source to drive scalable, automated catalyst design.

1 Introduction

Achieving high selectivity and durability in heterogeneous catalyst systems represents a key objective in industrial refining and specialty chemical synthesis. In gas streams containing target alkenes alongside minor fractions of terminal alkynes, active transition metal facets are often highly vulnerable to rapid deactivation. Trace amounts of terminal alkynes coordinate strongly to active metal sites, poisoning the catalyst and rendering it unreactive toward primary conversion streams.

Evaluating these deactivation pathways using conventional electronic structure modeling presents severe operational challenges. Standard density functional theory (DFT) computations require massive iterative solutions to the Kohn-Sham equations, making it impossible to perform high-throughput screening over vast structural libraries. Furthermore, standard single-component descriptors, like the localized d-band center or static adsorption energies, fail to separate different adsorption mechanisms. They routinely confuse harmless, temporary surface interactions with permanent, destructive chemical transformations.

To solve this operational bottleneck, this paper presents Project Descriptometrix. This architecture bridges the gap between quantum mechanics and automated material discovery. By projecting the interaction space onto a fast, non-iterative 3×3 tridiagonal matrix representation, we capture complex electronic states in fractions of a millisecond. This framework uncovers hidden materials that break traditional linear scaling laws, offering a comprehensive, deployable solution for automated catalyst discovery.

2 Mathematical Formulation & Matrix Core

The foundation of our screening engine relies on an analytical matrix representation that models a three-center system comprising a localized metal active site (M) and the two carbon centers (C_1, C_2) of an approaching terminal alkyne. We write the localized tridiagonal Hamiltonian operator matrix as:

$$H = \begin{pmatrix} \alpha_M & \beta_{M\text{-}C1}(r) & 0 \\ \beta_{M\text{-}C1}(r) & \alpha_{C1} & \beta_{CC} \\ 0 & \beta_{CC} & \alpha_{C2} \end{pmatrix} \tag{1}$$

Where α_M, α_{C1}, and α_{C2} represent the isolated spatial site energies, and β_{CC} defines the internal carbon-carbon triple bond coupling. The interfacial interaction term $\beta_{M\text{-}C1}(r)$ models the spatial distance (r) using a Wolfsberg-Helmholz approximation coupled to an exponential decay envelope:

$$\beta_{M\text{-}C1}(r) = 1.75 \times \exp[-\kappa(r - r_0)] \times \left(\frac{\alpha_M + \alpha_{C1}}{2}\right) \tag{2}$$

Solving this tridiagonal system yields exact eigenvalues (ϵ_i) and corresponding eigenvectors (v_i). From these analytical vectors, we derive two non-iterative descriptors that govern the poisoning pathway:

1. **Backbonding Susceptibility (ΔE_π):** This metric evaluates the electronic charge donation into the alkyne's anti-bonding orbitals:

$$\Delta E_\pi = \frac{\beta^2_{M\text{-}C1}}{\alpha_M - (\alpha_{C1} - 0.5)} \tag{3}$$

2. **Coulson Terminal Charge Redistribution (Δq_{C1}):** This parameter quantifies the local electronic displacement directly on the terminal carbon atom:

$$\Delta q_{C1} = 1.0 - 2.0 \cdot (v_{1,0})^2 \tag{4}$$

3 Results and Discussion

3.1 High-Throughput Computational Validation

The analytical Descriptometrix engine was validated by tracking its performance across a diverse verification library. As summarized in Table 1, the framework accurately mirrors the trends found in computationally demanding first-principles benchmarks while completing its evaluations in a fraction of a millisecond.

3.2 Decoupling Reversible Coordination from Permanent Deactivation

By plotting the candidate library across a two-dimensional phase map defined by our dual matrix descriptors, we successfully separate benign coordination from destructive catalyst poisoning. Traditional descriptor models tend to bundle these two interactions together, leading to high error rates.

Our analysis isolates a distinct electronic boundary. When the backbonding parameter (ΔE_π) exceeds 0.15 eV and the terminal carbon charge redistribution (Δq_{C1}) drops below -0.05 e, the

Table 1: Material library profiles, calculated analytical matrix descriptors, predicted kinetic abstraction barriers (E_a), and mapped mechanical safety domains.

Catalyst System	**Alkyne Intermediary**	$\mathbf{\Delta E_\pi}$ (eV)	$\mathbf{\Delta q_{C1}}$ (e)	**Pred.** E_a **(eV)**	**Mechanistic St**
Pd(111) Facet	Acetylene (Baseline)	0.1834	−0.1245	0.767	Critical Poisonin
Pd(111) Facet	Propyne (EDG)	0.1621	−0.0984	0.868	Critical Poisonin
Pd(111) Facet	Trifluoromethylacetylene (EWG)	0.2215	−0.1841	0.565	Critical Poisonin
Au(111) Facet	Acetylene (Baseline)	0.0412	+0.0021	1.347	Safe (Reversible)
Fe_1–N_4 SAC	Acetylene (Baseline)	0.0104	−0.0012	1.422	Safe (Spatial Co
Pd_1/Ag(111) SAA	Acetylene (Baseline)	0.0911	+0.0742	1.354	Safe (Charge Inv
Custom Alloy Site	Acetylene (Baseline)	0.1104	−0.0115	1.154	Safe Configuratio

system undergoes an irreversible transition. This high negative charge concentration polarizes the localized $\equiv$ C–H bond, severely lowering the activation energy (E_a) required for surface proton abstraction. Once the proton leaves, the alkyne forms an unreactive, covalent σ-metal bond that permanently deactivates the catalyst site. Systems that remain outside this critical electronic boundary avoid this trap, allowing the alkyne to desorb harmlessly and keep the active site open for continuous alkene processing.

3.3 Bypassing Linear Scaling Relations via Structural Anomalies

The primary hurdle in engineering poisoning-resistant surfaces is breaking the linear scaling relations that inherently couple the binding energy of target reactants to those of surface contaminants. Our high-throughput screening unmasked two distinct structural anomalies that successfully bypass these performance limitations.

The first class features spatial confinement anomalies, as seen in the single-atom catalyst Fe_1–N_4. While classical models assume isolated iron atoms are highly reactive and vulnerable, the local square-planar (D_{4h}) symmetry restricts the metal's spatial degrees of freedom. This structural locking prevents the geometric distortions needed for backbonding, driving ΔE_π down to a negligible 0.0104 eV and ensuring high poisoning resistance.

The second class relies on dynamic charge inversion, typified by the single-atom alloy Pd_1/Ag(111). Local differences in electronegativity drive electron density from the surrounding silver host matrix toward the isolated palladium single atom. This local electronic accumulation completely flips the polarization vector, forcing Δq_{C1} to become positive ($+0.0742\ e$). This positive charge creates an electrostatic repulsion against the terminal alkyne proton, keeping the active site pristine and selectively open for target alkene conversion.

4 Industrial Scaling & Production-Ready Deployment

To evaluate these materials under realistic industrial operations, our kinetic parameters were integrated into a stochastic Kinetic Monte Carlo (kMC) simulation. The model tracked catalyst site availability over extended process timelines featuring a continuous alkene feed mixed with 500 ppm alkyne poison.

The time-evolution profiles underscore the vulnerability of traditional materials. Pristine Pd(111) surfaces experienced a rapid **90% loss of available active sites within 1.5 hours** due to fast, uninhibited σ-acetylide accumulation. In contrast, our engineered Pd_1/Ag(111) single-atom alloy maintained over **98.5% active site availability** across a simulated week-long continuous run, proving that suppressing the proton abstraction pathway translates directly into long-term industrial durability.

To maximize open-science accessibility and enable seamless integration into automated experimental loops, the complete screening framework has been packaged into a lightweight, containerized REST API using FastAPI, allowing researchers to evaluate custom materials in real time.

5 Conclusion

Project Descriptometrix demonstrates a successful strategy for bypassing slow, resource-heavy quantum chemical calculations by deploying physics-informed analytical matrices. The framework successfully decouples harmless, temporary surface interactions from permanent chemical deactivation channels on a two-dimensional phase map. By identifying unique structural anomalies like spatial confinement matrices and charge-inversion single-atom alloys, the engine provides a reliable, high-speed approach for discovering durable, poisoning-resistant catalysts for industrial chemical streams.